How multiplayer works

Up to 10 explorers share one world, one riddle and one race to the well. One player's laptop runs the game logic and the AI, and a free Cloudflare relay connects everyone, wherever they are.



Host (Srujan) · connected through the live Cloudflare relay

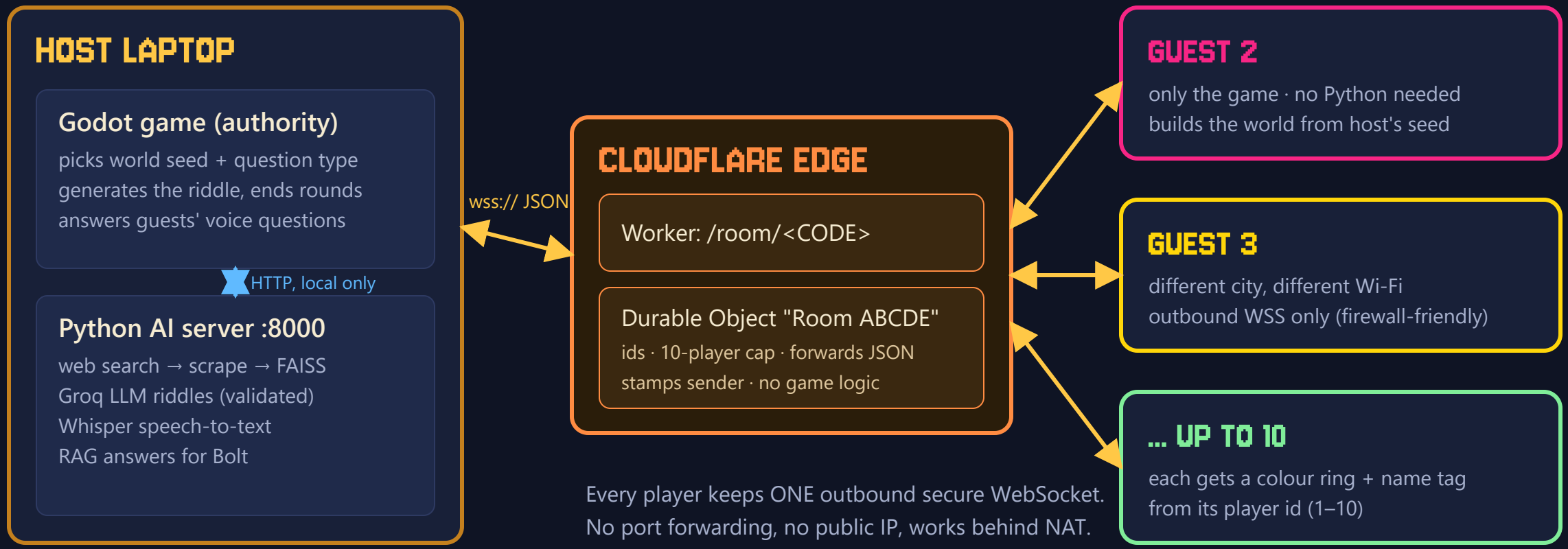


Guest (Arjun) on another instance · same world, same hint, same chat, same scoreboard

THE BIG PICTURE

One host, one relay, many players

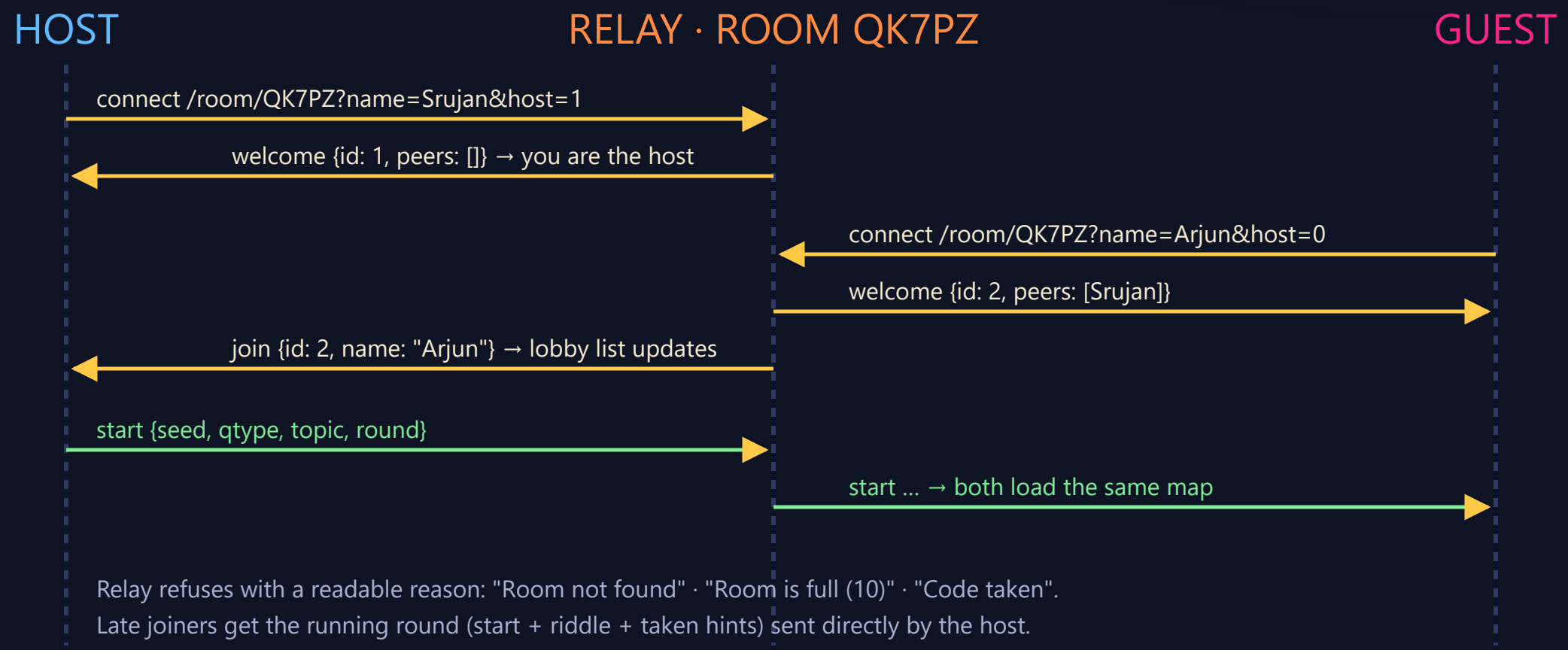
The relay is deliberately simple: it only forwards messages. The host's laptop does the real work (world seed, riddle generation, round flow), as in a LAN game, without anyone touching router settings.



JOINING A ROOM

Five letters and you're in

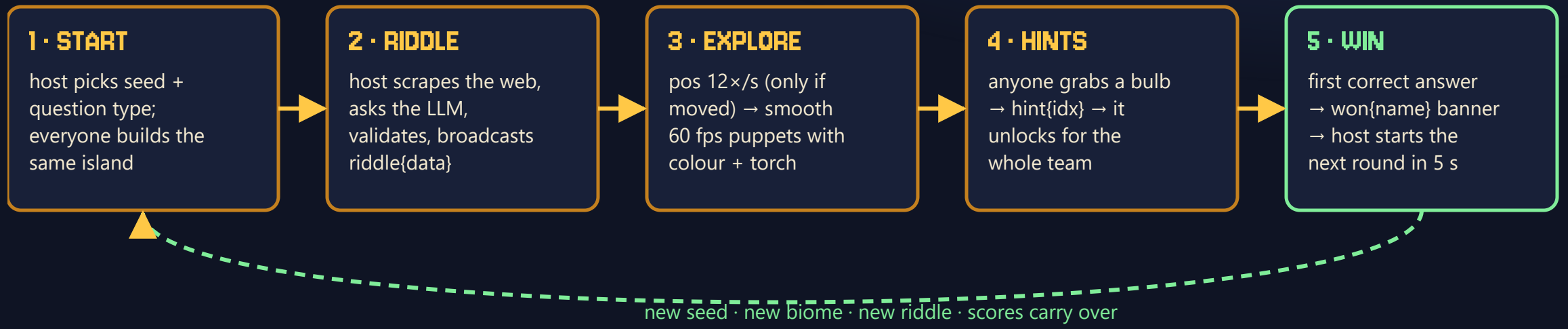
The host's game makes a 5-letter code (no confusable 0/O/1/I). The relay names one Durable Object after it, so everyone with the code lands in the same room on Cloudflare's network.



Relay refuses with a readable reason: "Room not found" · "Room is full (10)" · "Code taken".
Late joiners get the running round (start + riddle + taken hints) sent directly by the host.

ONE ROUND, STEP BY STEP

Shared riddle, shared hints, first answer wins



Answering stays personal

Each player opens a well and answers in their own popup (MCQ, Wordle, Whack, KBC…). A wrong answer costs only **your** hearts.

Nobody gets kicked

Lose all hearts online and you respawn after 3 s with a −15 score penalty. The round goes on.

Leaving is clean

Guest leaves → "X left" banner. Host leaves → relay closes the room and everyone returns home with a message.

WHY EVERY SCREEN SHOWS THE SAME ISLAND

Send a seed, not a map

The island is 280×280 tiles plus wells, trees and flowers, roughly 80,000 cells. Sending all of that would be slow. Instead the host sends **one number**.

- `seed(round.seed)` runs before terrain, biome, wells, trees, flowers and the spawn pad, so the same code with the same seed builds the same world.
- Hint bulbs are placed after the riddle arrives with `seed + 1`, so every player sees them in the same spots.
- Then `randomize()` again, so cosmetic things (birds, rain, flicker) stay lively on each screen.
- Cost: **~60 bytes** to start a round instead of megabytes.

1
number syncs the world

12/s
position updates, only when moving

~1/s
heartbeat when idle



Guest view: the host is the blue ring, the guest the pink one. Terrain, trees and wells match the host's screen exactly; the chat bubble, the shared hint (1/6) and both torches came over the relay, and the two torches share one light instead of glaring.

WHAT TRAVELS OVER THE WIRE

Thirteen small JSON messages

Type	Who sends	Fields	When
`welcome`	relay → joiner	id, peers[]	on connect
`join` / `leave`	relay → room	id, name	someone arrives / drops
`error` / `host_left`	relay → player	msg	full room, bad code, host quit
`start`	host → all	seed, qtype, topic, round	new round
`riddle`	host → all	riddle, options, solution, hints, fact	once per round
`pos`	everyone	x, y, anim, frame, flip, score, torch	12 Hz while moving, ~1 Hz idle
`hint`	whoever grabbed it	idx	bulb collected
`chat`	anyone	text (≤120 chars)	press T
`won`	the winner	name, answer	correct answer
`api` / `api_reply`	guest ⇄ host	path, base64 body / code, body	guest talks to Bolt

The relay overwrites `from` on every message, so a client can't pretend to be the host. Messages over 768 KB are dropped (enough for a 12-second voice clip).

VOICE FOR GUESTS

The host's laptop is everyone's AI

Guests can talk to Bolt without installing Python: their audio rides the relay to the host, whose server transcribes and answers it.

GUEST MIC
hold V or hands-free (VAD) → 16 kHz WAV

→

RELAY
api{path: /api/transcribe, b64 WAV} → host

→

HOST GAME
checks path is allow-listed, then calls 127.0.0.1:8000

→

AI SERVER
Whisper → text
RAG + "never spoil the answer" prompt

→

GUEST BOLT
api_reply → speech bubble + text-to-speech voice

Measured locally: Whisper transcription 0.47 s · Bolt's answer ≈ 1 s · a 30 s timeout falls back to an offline tip.

Two hops (transcribe, then ask) keep each message small and reuse the exact endpoints the host itself uses.



Tested end-to-end

A spoken question ("Hey Bolt, what is a tuple in Python?") was fed into a guest instance. The host's server logged `/api/transcribe 200` then `/api/rag 200`, and the guest's Bolt displayed and spoke the answer.

RUNNING IT FOR FREE

A free Cloudflare account is enough

What's deployed

`wss://endless-worlds-relay.srujanmpadmashali.workers.dev`

- Worker + SQLite-backed Durable Object, both available on the **Workers Free** plan
- WebSocket **Hibernation API**: an idle room costs nothing while sockets stay open
- Incoming WebSocket messages are billed at **20 : 1** against the request quota, and outgoing ones are free

Back-of-envelope traffic

Room	msgs / s (moving)	billed req / hour
2 players	24	≈ 4,300
5 players	60	≈ 10,800
10 players	120	≈ 21,600

Worst case: everyone moves non-stop at 12 Hz, ÷20 billing ratio. Standing still drops to ~1 msg/s per player. Check Cloudflare's current free daily request quota on the Durable Objects limits page; going over it fails requests for the rest of the day and never bills you.

Deploy in two commands

`cd relay` → `npx wrangler login` → `npm run deploy`, then put `RELAY_URL=wss://…workers.dev` in `.env`.

ROBUSTNESS & TESTING

What was checked

Relay test suite

- room-not-found for guests without a host
- host gets id 1, guests 2…10
- broadcast + sender stamping (spoofed `from` ignored)
- direct messages reach only their target
- 11th player rejected: "Room is full"
- leave + host_left delivered

Passed locally (wrangler dev) and on the live worker.

Game-level runs

- 3 instances in one room, AI riddle shared
- hint grabbed by host appears taken for guests
- guest wins → "You solved it" / "Arjun solved it"
- host auto-starts round 2 for everyone
- late joiner catches up mid-round
- two **packaged .exe** builds over the live relay

Known limits

- Host quits → room ends (no host migration)
- Only the host needs the AI server; without it, the offline question bank is used
- Builds carry only the relay/API URLs; API keys stay with the host's AI server

FOR PLAYERS

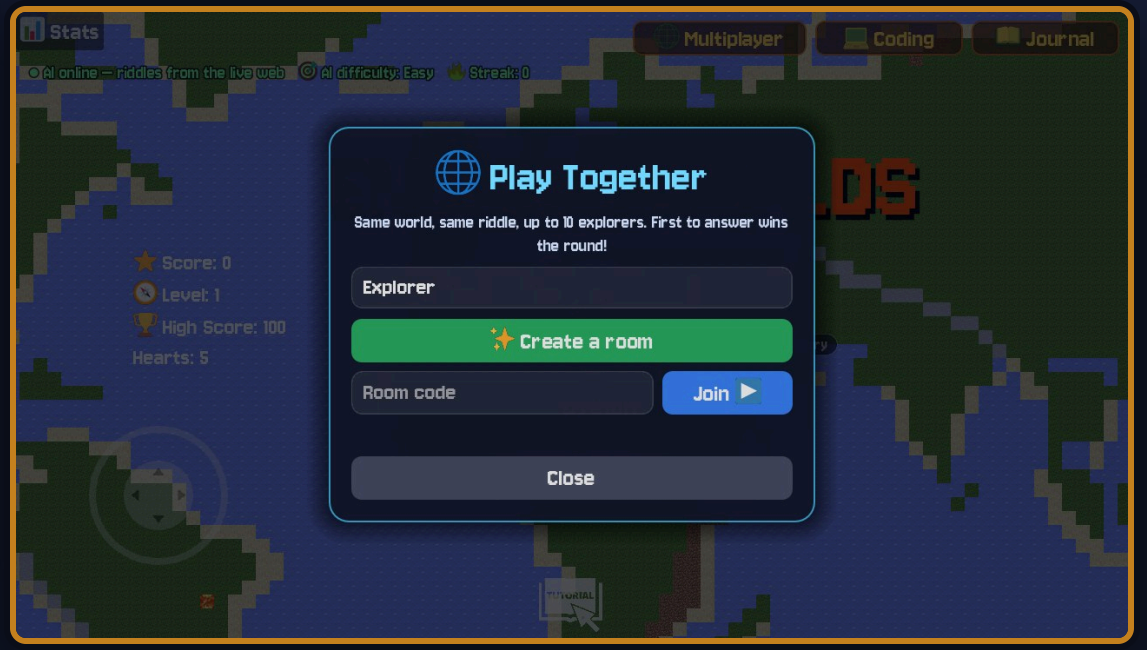
Play together in 60 seconds

Host

1. (Optional) run `start_server.bat` for live AI riddles + voice
2. Open the game → **Multiplayer** → type your name → **Create room**
3. The code is copied: send it to friends
4. Press **Start** when everyone is listed

Guests

1. Unzip and open `EndlessWorlds.exe`
2. **Multiplayer** → name → type the code → **Join**
3. Wait for the host to start, then race to the well!



The lobby: create or join, live player list with colours

WASD move | V talk | T chat | F torch | Esc pause